sPHENIX isolated photons on the world xT-scaling curve

PPG12 — isolated photon cross section in p+p at $\sqrt{s} = 200$ GeV

2026-06-02

Abstract

We place the PPG12 sPHENIX isolated-photon measurement on the world x_{T}-scaling curve of direct-photon cross sections, reproducing Fig. 1 (left) of arXiv:1901.10950 (Bock, Hard Probes 2018). After scaling by $(\sqrt{s})^{4.5}$, the sPHENIX points fall on the universal curve and lie at or just below the PHENIX 200 GeV direct-photon data, consistent within combined uncertainties. This is an external, cross-experiment check on the absolute normalization of the PPG12 cross section against 17 measurements spanning $\sqrt{s} = 19.4$ GeV to 13 TeV.

1 What Figure 1 is, and why the overlay is meaningful

x_{T} scaling exploits the leading-order pQCD expectation that the invariant cross section for a hard $2 \to 2$ process factorizes as $E\,\mathrm{d}^3\sigma/\mathrm{d}p^3 = (1/\sqrt{s})^n\,F(x_{\mathrm{T}})$, with n governed by the effective parton–parton scattering and the PDF/fragmentation evolution. Multiplying every measurement by $(\sqrt{s})^n$ collapses the different beam energies onto a single curve $F(x_{\mathrm{T}})$. The PHENIX 200 GeV direct-photon paper (PRD86 072008) made exactly this figure, and the Bock proceedings is a later, broader rendering of the same compilation. Because the compilation already contains a 200 GeV entry (PHENIX), adding the PPG12 200 GeV point is a direct apples-to-apples comparison at fixed $\sqrt{s}$, plus a test that our spectral shape tracks $F(x_{\mathrm{T}})$.

PPG12 measures *isolated* photons ($E_{\mathrm{T}}^{\mathrm{iso}}$, $R = 0.3 < 4$ GeV truth definition). The compilation mixes isolated (LHC, SppS, Tevatron, ISR) and inclusive-direct (PHENIX, E706, NA24, WA70, UA6) measurements, as the figure caption signals with the parenthetical "(isolated) direct photon". To make the same-energy comparison explicit we plot PHENIX both as published (inclusive-direct) and multiplied by its own measured isolated/inclusive-direct ratio (PRD86 Fig. 13a). PPG12 sits at or just below the lower (isolated) of the two PHENIX curves.

2 Data provenance

HEPData is the natural source, but its download endpoints are behind a Cloudflare managed challenge that cannot be solved from the analysis cluster. The numbers HEPData serves are digitized from the published papers, and arXiv e-print sources are not blocked, so we pulled the identical data tables straight from the papers' own LaTeX sources. Each external extraction was performed by one agent and independently re-extracted and checked point-by-point by a second.

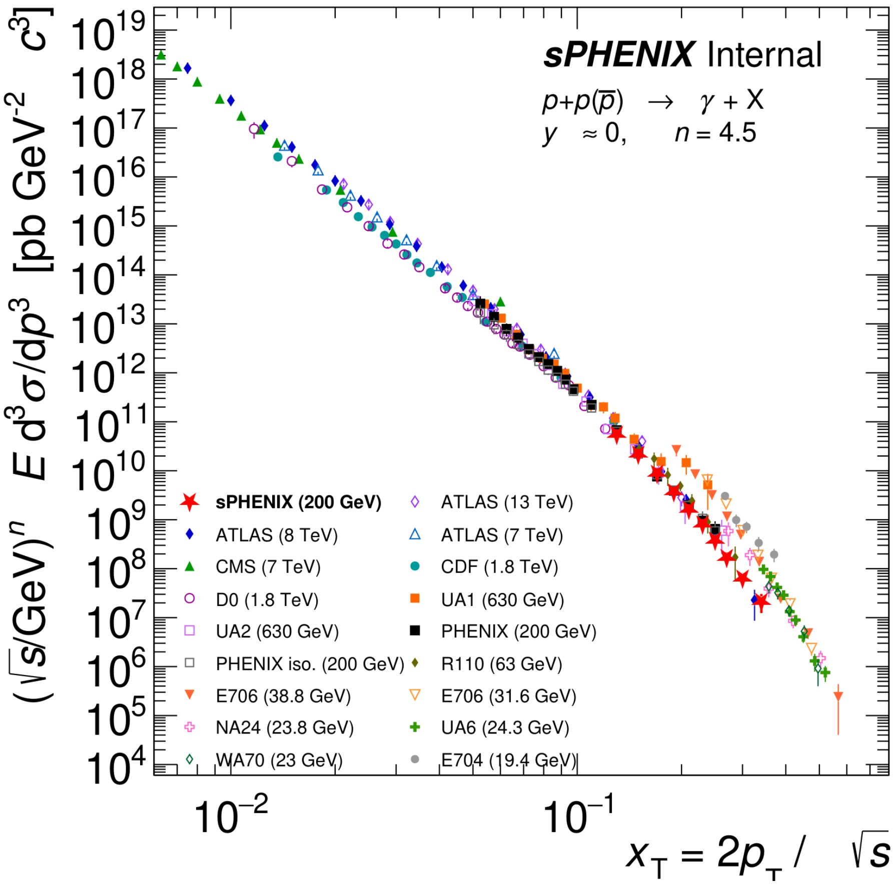

Figure 1: $(\sqrt{s}/\mathrm{GeV})^n\, E\, \mathrm{d}^3\sigma/\mathrm{d}p^3$ (units $\mathrm{pb\,GeV^{-2}}c^3$, $n = 4.5$) versus $x_\mathrm{T} = 2p_\mathrm{T}/\sqrt{s}$ at $y \approx 0$ for $p + p(\bar{p}) \to \gamma + X$. The PPG12 sPHENIX measurement (red stars) is overlaid on the world compilation.

Experiment	$\sqrt{s}$ [GeV]	Reference	observable	η range	iso	pts
sPHENIX (this)	200	PPG12 `Photon_final`	$\mathrm{d}\sigma/\mathrm{d}E_\mathrm{T}$	$\lvert\eta\rvert < 0.7$	yes	10
ATLAS	13000	PLB770 473	$\mathrm{d}\sigma/\mathrm{d}E_\mathrm{T}$	$\lvert\eta\rvert < 0.6$	yes	14
ATLAS	8000	JHEP08(2016)005	$\mathrm{d}\sigma/\mathrm{d}E_\mathrm{T}$	$\lvert\eta\rvert < 0.6$	yes	22
CMS	7000	PRL106 082001	$\mathrm{d}\sigma/\mathrm{d}E_\mathrm{T}$	$\lvert\eta\rvert < 1.45$	yes	11
ATLAS	7000	PLB706 150	$\mathrm{d}\sigma/\mathrm{d}E_\mathrm{T}$	$\lvert\eta\rvert < 0.6$	yes	8
CDF	1800	PRL73 2662	$\mathrm{d}^2\sigma/\mathrm{d}E_\mathrm{T}\,\mathrm{d}\eta$	$\lvert\eta\rvert < 0.9$	yes	16
D0	1800	PRL77 5011	$\mathrm{d}^2\sigma/\mathrm{d}E_\mathrm{T}\,\mathrm{d}\eta$	$\lvert\eta\rvert < 0.9$	yes	23
UA1	630	PLB209 385	$E\,\mathrm{d}^3\sigma/\mathrm{d}p^3$	$\lvert\eta\rvert < 0.8$	yes	15
UA2	630	PLB288 386	$E\,\mathrm{d}^3\sigma/\mathrm{d}p^3$	$\lvert\eta\rvert < 0.76$	yes	13
PHENIX	200	PRD86 072008	$E\,\mathrm{d}^3\sigma/\mathrm{d}p^3$	$\lvert\eta\rvert < 0.25$	no	18
PHENIX (isolated)	200	PRD86 Fig.10×13a	$E\,\mathrm{d}^3\sigma/\mathrm{d}p^3 \times R$	$\lvert\eta\rvert < 0.25$	der.	18
R110 (CMOR)	63	NPB327 541	$E\,\mathrm{d}^3\sigma/\mathrm{d}p^3$	$\lvert\eta\rvert < 0.8$	yes	7
E706	38.8	PRD73 032004	$E\,\mathrm{d}^3\sigma/\mathrm{d}p^3$	$-1.0<y<0.5$	no	9
E706	31.6	hep-ex/0407011	$E\,\mathrm{d}^3\sigma/\mathrm{d}p^3$	$\lvert y\rvert < 0.75$	no	7
NA24	23.8	PRD36 8	$E\,\mathrm{d}^3\sigma/\mathrm{d}p^3$	$-0.65<\eta<0.52$	no	5
UA6	24.3	PLB436 222	$E\,\mathrm{d}^3\sigma/\mathrm{d}p^3$	$-0.1<\eta<0.9$	no	9
WA70	23.0	ZPC38 371	$E\,\mathrm{d}^3\sigma/\mathrm{d}p^3$	$\lvert x_F\rvert < 0.05$	no	5
E704	19.4	PLB345 569	$E\,\mathrm{d}^3\sigma/\mathrm{d}p^3$	$\lvert x_F\rvert < 0.15$	yes	5

Most of the pre-arXiv SppS, Tevatron, ISR and fixed-target entries were recovered directly: CDF, D0, UA2, NA24 and UA6 from INSPIRE-hosted full-text PDFs (typeset tables read with PyMuPDF, not digitized from figures), UA1 and R110 from the Wayback-archived Durham HepData, and E704 and WA70 from theory-paper tabulations. Each was independently re-extracted, unit-checked, and validated against the universal band. Still missing (HEPData-only, no reachable machine-readable copy): the ISR sets R806, R807 and R108, and the two older PHENIX 200 GeV papers — all in x_T regions already covered, so they do not change the comparison.

3 Conversion to the common observable

Each spectrum is reduced to the invariant cross section at midrapidity and then x_T-scaled. For an invariant cross section (PHENIX, UA1, UA2, R110, E706, E704, NA24, WA70, UA6), $I \equiv E\,\mathrm{d}^3\sigma/\mathrm{d}p^3$ is used as published (the R110 and NA24 CGS cm^2 units are converted to pb). For a differential cross section,

$$I = \frac{1}{\Delta\eta}\,\frac{\mathrm{d}\sigma}{\mathrm{d}E_\mathrm{T}}\,\frac{1}{2\pi p_\mathrm{T}},$$

with $\Delta\eta$ the central η-bin width when the cross section is η-integrated (1.2 for ATLAS $|\eta| < 0.6$, 2.9 for CMS $|\eta| < 1.45$, 1.4 for PPG12 $|\eta| < 0.7$) and $\Delta\eta = 1$ for CDF and D0, which are already quoted per unit pseudorapidity. Cross-section units are normalized to pb. Then $Y = (\sqrt{s})^{4.5}\,I$ and $x_\mathrm{T} = 2p_\mathrm{T}/\sqrt{s}$. Following the PHENIX compilation [1], data points consistent with zero (lower error bar reaching zero) are excluded from the plot; this removes one near-threshold point each from E706 (31.6 GeV) and UA1.

The strongest validation is internal and now spans eight independent collider and RHIC measurements — CDF and D0 (1.8 TeV), UA1 and UA2 (630 GeV), ATLAS (7, 8, 13 TeV), CMS (7 TeV), and PHENIX and PPG12 (200 GeV). All collapse to within $\sim$2–3$\times$ at every shared x_T from 0.014 to 0.13, across a factor 65 in $\sqrt{s}$. A wrong $\Delta\eta$, 2π, cm^2$\to$pb, or unit factor in any one of them would be far larger than this residual scatter. CMS sits up to $\sim$2$\times$ above the others in its overlap, consistent with centroid bias in its very wide E_T bins.

4 Findings

1. The sPHENIX points lie on the universal x_T curve over the reported range $x_\mathrm{T} \approx 0.13$–$0.34$ ($E_\mathrm{T} = 12$–36 GeV). The $E_\mathrm{T} < 12$ GeV trigger-turn-on bins and the overflow bin are excluded, matching the range PPG12 reports.

2. **Same energy:** at $x_\mathrm{T} = 0.13$, PPG12/PHENIX(incl.-direct) = 0.82 and PPG12/PHENIX(isolated) = 0.87. The deficit is expected (isolation removes a fragmentation component) and is inside the combined uncertainties (PPG12 $\approx$14–21%, PHENIX $\approx$16–20% in this overlap region).

3. **Cross-energy** scaling holds at the factor-$\sim$2–3 level across $\sqrt{s} = 200$ GeV to 13 TeV: the eight collider+RHIC datasets agree to $\sim$2–3$\times$ over $x_\mathrm{T} = 0.014$–0.13 (at $x_\mathrm{T} = 0.13$ the full spread is 2.2$\times$; PHENIX–PPG12 1.2$\times$). Part of this spread is the isolated-vs-inclusive-direct mixing, not pure x_T-scaling violation.

4. The low-$\sqrt{s}$ fixed-target and ISR measurements (E704 19.4, NA24 23.8, WA70 23, E706 31.6/38.8, R110 63 GeV) sit progressively above the band at high x_T (a factor of several up to $\sim$15$\times$ by $x_\mathrm{T} \approx 0.3$) — the well-known breakdown of single-n x_T scaling at low $\sqrt{s}$. The effective power $n_\mathrm{eff}(x_\mathrm{T}, \sqrt{s})$ is not universal but grows away from the naive value 4 through

QCD scale-breaking [4], and the low-$\sqrt{s}$ cross section is further enhanced near partonic threshold (soft-gluon resummation [7]; the historical k_T interpretation [8]). PHENIX [1] flag the same "exception at low $\sqrt{s}$" in the original compilation. The effect reproduces consistently across all five experiments and matches Bock's figure (compressed there on its ∼37-decade axis).

The headline for the analysis: the PPG12 absolute normalization is consistent with the world data once placed on the x_T-scaling curve, and in particular with the only other midrapidity direct-photon measurement at 200 GeV.

5 Verification

The numerical re-derivation agent recomputed every quoted number from the ROOT file and raw tables from scratch: PPG12 ($E_\mathrm{T} = 13, 25$ GeV from `h_unfold_sub_result`), PHENIX, and ATLAS conversions all reproduce to floating-point (MATCH, 0.0000%). The figure passed the plot-cosmetics review. The "within a factor of 2" at $x_\mathrm{T} = 0.13$ is literally 2.15× when the highest LHC point is compared to the lowest RHIC point, so the statement is meant in the round sense.

References

1. PHENIX Collaboration, A. Adare *et al.*, "Direct-photon production in $p + p$ collisions at $\sqrt{s} = 200$ GeV at midrapidity," Phys. Rev. D **86** (2012) 072008 [arXiv:1205.5533]. *Origin of this x_T figure; $n_\mathrm{eff} = 4.5$; the "exclude points consistent with zero" convention; the low-$\sqrt{s}$ exception.*

2. F. Bock, "Shining a light on the QGP — electroweak probes experimental summary," arXiv:1901.10950 (Hard Probes 2018). *The figure reproduced here.*

3. R. Blankenbecler, S. J. Brodsky, J. F. Gunion, Phys. Lett. B **42** (1972) 461. *x_T scaling.*

4. R. F. Cahalan, K. A. Geer, J. B. Kogut, L. Susskind, Phys. Rev. D **11** (1975) 1199. $n_\mathrm{eff}(x_T, \sqrt{s})$ *from QCD scale-breaking.*

5. W. Vogelsang, M. R. Whalley, J. Phys. G **23** (1997) A1. *World direct-photon data compilation (source for WA70, E704).*

6. P. Aurenche, J. P. Guillet, E. Pilon, M. Werlen, M. Fontannaz, Phys. Rev. D **73** (2006) 094007 [arXiv:hep-ph/0602133]. *NLO pQCD critical study.*

7. D. de Florian, W. Vogelsang, Phys. Rev. D **72** (2005) [arXiv:hep-ph/0506150]. *Threshold (soft-gluon) resummation — low-$\sqrt{s}$ enhancement.*

8. L. Apanasevich *et al.*, Phys. Rev. D **59** (1999) [arXiv:hep-ph/9808467]. *k_T-smearing interpretation of the fixed-target excess.*

Reproduce

```
cd plotting
python3 xtscaling/convert.py
root -l -b -q 'xtscaling/plot_xtscaling.C'
```

Code and data live under `plotting/xtscaling/` (`convert.py`, `plot_xtscaling.C`, `data/`, `converted/`); the figure is `plotting/figures/xtscaling_compilation.pdf`.